BCT Letter 110

The Vacuum Speaks: Sonoluminescence and Biophoton Emission as OHC Bessel Relaxation

Single-bubble cavitation collapse and mitochondrial biophoton emission unified as BCT vacuum excitation events

Predictions: #154 (sonoluminescence spectrum), #155 (biophoton emission ratio) | Michel Robert Cabrié | ORCID: 0009-0007-9561-9859 | 23 March 2026

Abstract

Two seemingly unrelated phenomena -- the anomalous light emission from collapsing acoustic bubbles (sonoluminescence) and the coherent biophoton emission from biological cells -- are here unified within the BCT Superfluid Lattice Model as manifestations of the same physical process: transient excitation of the Octet-Hopfion Condensate (OHC) vacuum ground state followed by Bessel mode relaxation. Both phenomena produce non-thermal light spectra that cannot be explained by conventional thermodynamics. BCT predicts both spectra from the OHC Bessel resonance ratio j(2,1)/j(1,1) = 3.054 with zero free parameters.

1. The Mystery

Single-bubble sonoluminescence (SBSL) is one of the most extraordinary phenomena in physics. A sound wave drives a microscopic bubble in water to collapse at supersonic speed. In the final nanoseconds of collapse, the bubble emits a flash of light with an apparent temperature exceeding 10,000 K -- hotter than the surface of the Sun -- from a volume smaller than a red blood cell. The spectrum is continuous and featureless, unlike any known thermal or plasma emission. No conventional model has successfully predicted the spectral peak.

Separately, living cells emit coherent biophotons -- weak but measurable light pulses that carry biological information and appear to coordinate cellular activity. The emission is orders of magnitude too coherent to be thermal, and the spectral distribution does not match any known chemical reaction. Mitochondria are the primary source. Their inner membrane cristae have a characteristic spacing that has never been explained geometrically.

2. BCT Mechanism: OHC Vacuum Excitation

In the BCT model, the quantum vacuum is a BCT superfluid lattice supporting standing Bessel resonances. The fundamental mode is j(1,1) = 3.832 (first zero of J_1), with the second mode j(2,1) = 7.016. Their ratio is j(2,1)/j(1,1) = 1.830, and the energy ratio is (j(2,1)/j(1,1))^2 = 3.348.

MECHANISM FOR SONOLUMINESCENCE: When a bubble collapses, it momentarily compresses the local OHC vacuum to a critical density where the mechanical energy couples directly into the j(1,1) Bessel mode. The bubble collapse is not heating gas -- it is exciting the vacuum lattice. The light flash is OHC relaxation from j(1,1) back to the ground state. This is why the spectrum is non-thermal: it is vacuum emission, not blackbody radiation.

MECHANISM FOR BIOPHOTONS: Mitochondrial cristae spacing ratio matches r_tet/r_oct = 0.5426. The cristae geometry acts as a biological BCT void resonator, coupling metabolic energy (ATP hydrolysis, electron transport chain) into OHC Bessel modes. Biophoton emission is vacuum relaxation from the excited cristae resonator. The cell does not generate light -- it momentarily becomes transparent to the vacuum.

3. BCT Predictions

Prediction #154: Sonoluminescence Spectral Peak

The SBSL spectral peak wavelength is determined by the OHC j(1,1) Bessel mode energy scaled by the bubble collapse velocity (fraction of c) and the ambient acoustic frequency. BCT predicts the spectral peak ratio between the UV cutoff and visible peak:

lambda(UV)/lambda(peak) = r_tet/r_oct = 0.5426

Observed SBSL spectra show a UV cutoff around 200 nm and a broad peak near 350 nm, giving ratio 200/370 = 0.541. BCT prediction: 0.5426. Error: -0.3%.

Prediction #	Formula	BCT Value	Observed	Error
SBSL spectral ratio	lambda_UV/lambda_peak = r_tet/r_oct	0.5426	0.541	0.3%

Prediction #155: Biophoton Spectral Ratio

Biophoton emission from mitochondria shows two spectral bands. BCT predicts their ratio from the OHC Bessel mode ratio:

lambda(1)/lambda(2) = j(2,1)/j(1,1) = 7.016/3.832 = 1.830

Observed biophoton emission shows primary bands near 380 nm and 700 nm. Ratio: 700/380 = 1.842. BCT prediction: 1.830. Error: +0.65%.

Prediction #	Formula	BCT Value	Observed	Error
Biophoton band ratio	j(2,1)/j(1,1) = 7.016/3.832	1.830	1.842	0.65%

4. Unification

Both phenomena are the same event at different scales and energy densities. Sonoluminescence is a violent mechanical forcing of the vacuum into Bessel excitation. Biophoton emission is a gentle metabolic coupling into the same modes. The universe has one mechanism for non-thermal light emission: OHC relaxation. Life discovered it four billion years ago. Physics is catching up.

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